

Kuratowski Calculus Engine

Unified Code Manual

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Part of Modulo Synthesis / Fractal Entropic Geometrodynamics

*“The universe is strictly computable, finite, and structurally
homeostatic. The rest is just counting.”*

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Theory

Chapter Overview

- *The two axioms of Fractal Entropic Geometrodynamics.*
- *The Sole Principle of Interaction (microscopic law).*
- *Kuratowski's Theorem and the origin of matter, forces, and generations.*
- *The Three Geometries: Order, Matter, and Logic.*
- *How each physics concept maps to a code module and data structure.*

1.1 Axiom 1 — Causal Sets

Spacetime is not a continuous manifold but a **discrete partial order** — a directed acyclic graph (DAG) where each node is a spacetime event and each edge encodes causal influence: $A \rightarrow B$ means “ A can causally influence B .” The DAG is generated by Poisson sprinkling N points into a 4D causal diamond ($|t| + r \leq T/2$) with volume $V = \pi T^4/24$ and $T = (24N/\pi)^{1/4}$ giving fundamental density $\rho \approx 1$.

Axiom 1: Causal Sets (from Vol I)

Spacetime is a locally finite partially ordered set. The order relation encodes causal structure; the counting measure (number of elements in an Alexandrov set) encodes spacetime volume. All geometry is emergent from order + number.

1.2 Axiom 2 — Quantum Dynamics

Axiom 2: Quantum Dynamics: The Non-Commutative Postulate (from Vol II)

The geometry and quantum phases of the causal set are evaluated over a **non-commutative von Neumann algebra**. Macroscopic geometry is not fundamental; it is an emergent phenomenon dictated by the quantum Fisher Information Metric (Petz metric) defined over the density states of causal histories. Causal histories do not commute: if events A and B are spacelike separated, the order in which they are evaluated affects the quantum phase.

Together, Chentsov's Theorem (classical, macroscopic) and Petz's Theorem (quantum, microscopic) establish a single information-geometric foundation for all of physics. Gravity is the coarse-grained Fisher geometry at large N ; quantum mechanics is the fine-grained Petz geometry at small N .

1.3 The Sole Principle of Interaction

Axiom 3: The Sole Principle of Interaction (Microscopic Law)

The entirety of microscopic network dynamics is governed by a single mandate: **the preservation of unitarity** (probability conservation) across discrete, non-

commutative causal links:

$$\sum_{\text{paths}} P(i \rightarrow f) = 1$$

All emergent forces (gauge fields) and geometric structures (gravity) are **homeostatic mechanisms** enacted by the causal network to satisfy this unitary principle under the topological constraints of Kuratowski's Theorem.

1.4 Hasse Diagram

The **transitive reduction** of the causal DAG. If $A \rightarrow B \rightarrow C$, the direct edge $A \rightarrow C$ is redundant and is removed. The Hasse diagram retains only nearest-neighbour causal links — the skeleton of spacetime.

1.5 Causal Prism — The Fundamental Particle

CRITICAL: The transitive reduction in Phase 1 produces a *triangle-free* DAG. K_4 complete subgraphs (4 mutually connected events) **cannot exist** in such a graph — transitive reduction eliminates all triangles by definition. The correct fundamental particle is the **Causal Prism** ($K_{2,N}$ bipartite, $N \geq 3$).

A Causal Prism consists of:

- Two **poles**: past pole u (origin) and future pole v (destination)
- $N \geq 3$ **belly nodes** (intermediates w_i)
- Directed edges: $u \rightarrow w_i$ and $w_i \rightarrow v$ for all $i \in \{1 \dots N\}$
- The belly = $\text{children}(u) \cap \text{parents}(v)$

These topological defects:

- Anchor probability flow and define particle-like spectral signatures.
- Cannot be “untied” without breaking the graph's connectivity — they are topologically stable.
- Are the *maximal* bipartite structures permitted by the triangle-free constraint.

1.6 Threat Contraction — Colour Confinement

Theorem 4: Kuratowski's Theorem (The Topological UV Limit)

A finite graph G is planar if and only if it does not contain a topological minor of K_5 (the complete graph on 5 vertices) or $K_{3,3}$ (the complete bipartite graph on $3 + 3$ vertices):

$$K_5 \not\leq_T G_{UV} \quad \text{and} \quad K_{3,3} \not\leq_T G_{UV}$$

This is the absolute UV boundary of the theory.

When an external node is connected to **both poles** of a Causal Prism AND ≥ 2 intermediates, it threatens the topological stability and is **absorbed** (vertex contraction) into the higher-degree pole — a discrete analogue of colour confinement. The topology enforces that quarks cannot exist in isolation.

Key Result

The Strong Force is not a fundamental gauge field. It is a strict topological obstruction — the causal network's response to the Kuratowski UV limit — preventing forbidden topological minors in the ultraviolet.

1.7 Spectral Dimension — Gravity

The spectral dimension measures how quickly a random walker returns to its starting point after t steps:

$$d_S(t) = -2 \frac{d(\ln P)}{d(\ln t)}$$

On a smooth 4D manifold, $d_S \rightarrow 4$. On the Hasse graph:

- **Vacuum:** d_S diverges (boundary saturation of the unstructured bulk).
- **With Causal Prism matter:** d_S stabilises at a finite value — the topological knot traps probability flow.

1.8 Causal Flux — Electromagnetism

Directed walkers follow the arrow of time (cause $\rightarrow$ effect only). The transmission probability between particle generations measures electromagnetic-like interactions:

- **Attraction:** Gen1 $\rightarrow$ AntiGen1 (opposite charges).
- **Repulsion:** Gen1 $\rightarrow$ Gen1 (same charges).

1.9 Three Generations — The Kuratowski Limit

The Standard Model has three copies of matter (electron/muon/tau, up/charm/top). In FEG the answer is a strict topological obstruction:

1. UV gravity compresses $d_S \rightarrow 2$ (effectively 2D topology).
2. Kuratowski's Theorem forbids K_5 in planar graphs.
3. The transitive reduction produces a triangle-free DAG.
4. Causal Prisms ($K_{2,N}$ bipartite) are the fundamental particles.

Causal Prisms are classified by the **Vol I phase function** $\varphi(w) = \text{sign}(\text{bulk_momentum}[w])$.

For each prism P with intermediates $\{w_1, \dots, w_N\}$:

- $g(P) = |\{\varphi(w_i)\}|$ counts distinct phase values — the **generation number** (1, 2, or 3).
- $\Phi(P) = \sum_i \varphi(w_i)$ gives the **net phase** — matter if $\Phi \geq 0$, antimatter if $\Phi < 0$.

Gen1 = $g=1 \wedge \Phi \geq 0$, Gen2 = $g=2$, Gen3 = $g=3$, Anti1 = $g=1 \wedge \Phi < 0$. The constraint $g \leq 3$ (at most 3 distinct sign values: $-1, 0, +1$) enforces exactly 3 stable generations.

1.10 The Three Geometries

The theory rests on three geometric pillars, each discovered through the simulation engine:

Key Result

The Three Geometries of Fractal Entropic Geometrodynamics:

1. **Geometry of Order** (Vol I): *Transitive Reduction $\rightarrow$ Lorentz invariance. The Hasse diagram encodes the causal structure of spacetime; counting elements in an Alexandrov set reproduces the volume element. Code: `diamond.rs` builds the Hasse DAG.*
2. **Geometry of Matter** (Vol II): *Causal Prism $K_{2,N} \rightarrow$ mass hierarchy. Topological defects in the Hasse graph carry quantized topological mass N and are classified into exactly three generations by the Kuratowski limit. Code: `skyrmion.rs` detects prisms and classifies generations.*
3. **Geometry of Logic:** *$O(N)$ scaling $\Leftrightarrow$ physical locality. Bounded Hasse degree ($D \leq 15$) implies that every topological operation—prism detection, threat contraction, spectral walk—is local, yielding **provably linear** $O(N)$ computational complexity. This is not an engineering optimisation but a computational proof that the theory encodes physical locality as an information principle. Code: `2-hop`*

candidate search in skyrmion.rs; single-pass streaming in diamond.rs.

Theorem 5: $O(N)$ Locality Theorem

Let $\mathcal{H}$ be the Hasse diagram of N events sprinkled into a 4D causal diamond at density $\rho \approx 1$. Then:

1. Every Hasse link has bounded proper time: $\tau_{\text{link}} \leq 8$.
2. Hence every node has bounded Hasse degree: $D \leq D_{\text{max}} \approx 15$.
3. Causal Prism detection via 2-hop traversal is $O(N \cdot D_{\text{max}}^3) = O(N)$.
4. All other phases (edge generation, spectral walkers, output) are already $O(N)$ or $O(W \cdot t)$.

The entire simulation scales linearly with the number of spacetime events.

1.11 Physics $\rightarrow$ Code Dictionary

Physics Concept	Code Module	Data Structure
Spacetime	diamond.rs	DAG of [f64; 4] points
Causal link	diamond.rs	CSR edge (adj_head, adj_data)
Particle (matter)	skyrmion.rs	Causal Prism ($K_{2,N}$ bipartite)
Strong force	skyrmion.rs	Threat contraction merge_into[i]
Charge (bulk momentum)	diamond.rs	bulk_momentum: Vec<i32>
Particle generation	skyrmion.rs	Vol I phase: $g(P) = \{\varphi(w_i)\} $
Antimatter	skyrmion.rs	CPT conjugate: $g=1 \wedge \Phi(P)<0$
Gravity (geometry)	spectral.rs	Undirected walkers $\rightarrow P(t) \rightarrow d_S(t)$
Electromagnetism	spectral.rs	Directed walkers $\rightarrow$ transmission prob.
Locality ($O(N)$)	skyrmion.rs	2-hop traversal, bounded degree $D \leq 15$
Topology export	anim_export.rs	TopologySlice $\rightarrow$ bincode .slice file

Table 1.1: Mapping between physics concepts and code implementations.

Architecture

Chapter Overview

- The four-phase pipeline: *sprinkle* → *hasse* → *defect* → *walkers* → *csv*.
- Two execution paths: *in-memory* vs *streaming* (single-pass $O(N)$).
- Tier system: *eigendecomposition* vs *Monte Carlo*.

2.1 Data Flow

```
sprinkle(N) -> [f64;4] points
|
build_hasse() -> CSR adjacency (adj_head, adj_data) + bulk_momentum
|
apply_defect() -> Causal Prisms, K5 threat absorption, generation classification
|
+--[--export-slice]--> write_slice() -> .slice file (bincode, early exit)
|
run_walkers() -> P(t), dS(t) for vacuum/defect/generations/flux
|
write_csv() -> results.csv
```

2.2 Phase Diagram

Phase	Module	Physics
1	diamond.rs	Vacuum generation: Poisson sprinkling + Hasse diagram
2	skyrmion.rs	Kuratowski contraction: Causal Prism detection + K_5 absorption
3	spectral.rs	Spectral dimension: random walk return probability
4	output.rs	CSV serialisation of ensemble-averaged observables

Table 2.1: The four simulation phases.

2.3 Execution Paths

2.3.1 In-Memory

Selected by `--inmemory` flag or auto-selected when RAM is sufficient. All data structures live in RAM. Realisations run in parallel via Rayon with an auto-selected concurrency limit (4 threads, or 2 for $N > 10M$; overridable via `--threads`). Causal Prism detection uses 2-hop local candidate search: $O(N \cdot D^3)$ with $D \leq 15$.

2.3.2 Streaming

Selected by `--stream` flag or auto-selected when RAM is insufficient. Phase 1 uses `scan_edges_with_analysis()` to perform a **single-pass** scan: degree counting, core classification, and core-edge collection are merged into one streaming pass over the edge generator. Causal Prism detection then operates on the core-induced CSR subgraph using the same 2-hop local search as the in-memory path. RAM usage stays under ~ 1.3 GB per realisation even at $N = 100M$. Concurrency is auto-selected by CPU count (capped at 8; overridable via `--threads`).

Mode selection is automatic based on available system RAM (via `sysinfo`), configurable via `causal_set.toml`, or overridable with CLI flags.

2.4 Tier System

Tier	Condition	Algorithm	Complexity
Eigendecomp	$N \leq 3,000$ (default <code>--eigen-cutoff</code>)	Dense $S = D^{-1/2}AD^{-1/2}$ eigendecomp	$O(N^3)$
Monte Carlo	$N > 3,000$ (default <code>--eigen-cutoff</code>)	Lazy random walkers on CSR	$O(W \cdot t)$

Table 2.2: Spectral dimension computation tiers.

Module Reference: **diamond** — Phase 1

Vacuum Generation

- *Poisson sprinkling in a 4D causal diamond.*
- *Hasse diagram construction (transitive reduction).*
- *Two tiers: sparse A^2 reduction ($N \leq 15k$) and OCI geometric construction ($N > 15k$).*

3.1 Functions

3.1.1 `sprinkle(n, rng) → (Vec<[f64;4]>, f64)`

Rejection-samples n points inside $|t| + r \leq T/2$ with $T = (24N/\pi)^{1/4}$ giving density $\rho \approx 1$. The volume of the 4D causal diamond is $V = \pi T^4/24$. The diamond is the intersection of the future light cone of $(t = -T/2)$ and the past light cone of $(t = +T/2)$.

3.1.2 `build_hasse_sparse(pts) → (coords, adj_head, adj_data, momentum)`

Full causal closure followed by sparse A^2 reduction. Edge (i, j) is a Hasse link iff $A^2[i, j] = 0$ (no 2-hop path exists). Used for $N \leq 15,000$. Parallel over source nodes via Rayon.

3.1.3 `build_hasse_direct(pts) → (coords, adj_head, adj_data, momentum)`

Direct geometric construction using the **Occupied Cell Index (OCI)** — a spatial indexing paradigm that replaces naive geometric shell enumeration (96% empty cells) with per-time-layer lists of occupied cells, queried via binary search. Features:

- Points are quantised onto an integer lattice and sorted by cell index for cache-coherent traversal (GridPoint struct).
- Per-time-layer lists of occupied cells, sorted by X coordinate.
- Binary search for the active X-band within the light cone.
- **Topological cutoffs** (Phase 1 optimisations):
 - `MAX_CAUSAL_DEPTH = 4`: proper-time cutoff. Hasse links beyond $\Delta t > 4$ are exponentially suppressed at $\rho \approx 1$ (was 10).
 - `MAX_HASSE_DEGREE = 15`: degree cap (Valve 1) — a node with 15+ descendants has fully populated its causal shadow (was unbounded).
- **Alexandrov volume short-circuit (Valve 2)**: If every causal candidate in a Δt -shell is transitively blocked (non-empty Alexandrov set between u and v), further shells have strictly larger Alexandrov volume and are guaranteed blocked. Breaks immediately when degree ≥ 1 .
- **Adaptive dry-shell termination (Valve 3)**: 2 consecutive dry Δt -shells (no new Hasse links) + ≥ 2 existing descendants triggers cutoff.
- Hollow light-cone shell optimisation: skip deep interior points where proper time $> \text{MAX_CAUSAL_DEPTH}$ units.

3.1.4 `stream_edges_to_file(n_total, chunk_size, path, seed)`

Streaming variant that writes Hasse edges to disk using a sliding window, for N too large to fit in RAM. Includes the same three termination valves (degree cap, Alexandrov short-circuit, adaptive dry-shell) as the in-memory path. Output format: continuous stream of

(u32, u32) in little-endian. Both physics invariants are preserved: full light cone search and transitive reduction via locally cached children coordinates.

3.1.5 `sprinkle_chunk(n_expected, t_min, t_max, half_t, rng, start_idx)`

Poisson-sprinkle points into a time-slab $[t_{\min}, t_{\max}]$. Used internally by the streaming path to fill the sliding window buffer.

3.1.6 `scan_edges_sparse_ext(n_total, chunk_size, seed, on_batch)`

Extended streaming edge scanner. Identical to `scan_edges_sparse` except the callback receives the current `safe_threshold` as a second argument: `FnMut(&[(u32, u32)], f64)`. This enables callers to track node finalization (a node's total degree is known once `safe_threshold > tnode + safety_margin`).

3.1.7 `scan_edges_with_analysis(n_total, chunk_size, seed, core_num, core_den)`

Single-pass streaming analysis that merges degree counting, core classification, and core-edge collection into one scan over the edge generator. Returns `(in_deg, out_deg, is_core, directed_core_edges)`.

Pipeline inside the callback:

1. Update `in_deg[v]` and `out_deg[u]` for each edge.
2. Track finalized nodes via safe-threshold delay (`safe_threshold - 2 · max_dt - 4.0`).
3. Running degree histogram (10,000 bins) estimates the core threshold after 5% warmup.
4. Classify finalized nodes as core or non-core.
5. Collect edges where both endpoints are classified core; buffer edges where one endpoint is pending.
6. Final exact $O(N)$ histogram pass corrects any drift ($\pm 0.5\%$) and re-checks all collected core edges.

Key Result

The single-pass architecture eliminates the two-pass bottleneck (degree counting + core-edge collection), halving wall-clock time for $N > 1M$ streaming runs. At $N = 100M$ each realisation uses ~ 1.3 GB RAM ($13 \cdot N$ bytes).

3.2 Key Concepts

Definition 1: GridPoint

A spacetime event quantised onto an integer lattice for cache-coherent traversal. Fields: `p` (coordinates), `orig_idx`, `cell` (linearised 4D grid coordinate), `qt/qx/qy/qz` (quantised coordinates). Points are sorted by cell index so that spatially nearby events are contiguous in memory.

Definition 2: Occupied Cell Index (OCI)

Paradigm shift from geometric to spatial indexing. The naive approach iterates all cells in the light-cone shell for each source node. At large N the grid is sparse ($> 96\%$ cells empty), making geometric enumeration wasteful. OCI inverts the lookup:

1. Build per-time-layer lists of *occupied* cells only.
2. Sort each layer by X coordinate.
3. For each source, binary-search the X-band of the light cone within the occupied list, skipping all empty space.

This reduces the inner-loop cost from $O(\text{shell_volume})$ to $O(\text{occupied_cells})$, a $\sim 20\times$

speedup at $N > 100k$.

Module Reference: **skyrmion** — Phase 2

Kuratowski Contraction

- *Pure integer topology — zero floating-point.*
- *Core identification $\rightarrow$ Causal Prism ($K_{2,N}$ bipartite) detection $\rightarrow$ threat contraction $\rightarrow$ generation classification.*
- *Vol I phase classification: $g(P) = |\{\varphi(w_i)\}|$ (generation), $\Phi(P) = \sum \varphi(w_i)$ (matter/antimatter).*

4.1 Algorithm

CRITICAL: The transitive reduction produces a triangle-free DAG. K_4 complete subgraphs **cannot exist**. The fundamental particle is the **Causal Prism** ($K_{2,N}$ bipartite, $N \geq 3$): two poles with $N \geq 3$ belly nodes (intermediates satisfying $u \rightarrow w_i \rightarrow v$).

1. **Core selection:** Top 10% of nodes by undirected Hasse degree ($\approx$ the diamond's combinatorial centre).
2. **Causal Prism formation ($O(N)$ forward-forward 2-hop search): Minimum valence pruning:** nodes with out-degree $< \text{MIN_PRISM_SHARED}$ (3) are skipped before the 2-hop traversal, and candidate poles v with in-degree < 3 are skipped before the belly intersection. For each remaining core node u , collect 2-hop candidates by traversing $u \rightarrow w \rightarrow v$ through forward successors w (children of u) and their forward successors v (grandchildren of u , future pole candidates). Then count belly nodes $|\text{children}(u) \cap \text{parents}(v)|$ using the reverse CSR for $\text{parents}(v)$. If ≥ 3 , the pair forms a Prism candidate with the belly as intermediates. Greedy best-partner selection maximizes belly size per pole.

Complexity: $O(N_{\text{core}} \cdot D^3)$. With bounded Hasse degree $D \leq 15$: $O(10\text{M} \times 3,375) \approx 3.4 \times 10^{10}$ operations at $N = 100\text{M}$ — minutes instead of days.

Completeness proof: If u and v are poles of a timelike $K_{2,N}$, then every belly node w_i satisfies $u \rightarrow w_i \rightarrow v$. The forward-forward path $u \rightarrow w_i \rightarrow v$ guarantees v appears as a candidate of u . Every valid prism partner *must* have at least 3 belly nodes, each providing such a forward-forward path. The 2-hop search is therefore **complete**: no valid partner is missed. The valence guard is safe because any valid pole must have ≥ 3 children (belly nodes).

3. **Threat contraction:** Any external node connected to BOTH poles AND ≥ 2 intermediates is absorbed into the higher-degree pole (vertex contraction).
4. **Generation classification (Vol I phase):** Prisms classified by $\varphi(w) = \text{sign}(\text{bulk_momentum}[w])$. $g(P) = |\{\varphi(w_i)\}|$ gives the generation number (1, 2, or 3); $\Phi(P) = \sum \varphi(w_i)$ distinguishes matter ($\Phi \geq 0$) from antimatter ($\Phi < 0$).
5. **Transitive merge chain resolution:** Pointer chase to resolve chains ($O(N)$).
6. **Defect CSR construction:** Zero-copy CSR from merge map + pole-completion edges (one edge per Prism: origin $\leftrightarrow$ destination).

4.2 Functions

4.2.1 `apply_defect(n, adj_head, adj_data, bulk_momentum) → (DefectResult, TopologySummary, Vec<CausalPrism>)`

Main entry point. Finds Causal Prisms ($K_{2,N}$ bipartite structures with belly nodes), classifies them by topological signature into generations, performs threat contraction, and builds the defect CSR by adding pole-completion edges. The committed `Vec<CausalPrism>` is returned alongside the result so that callers (e.g. `--export-slice`) can access the raw prism geometry without re-detection.

4.2.2 `process_streaming(n_total, chunk_size, path, seed) → SpectralResult`

Out-of-core variant for streaming mode. Single-pass analysis via `scan_edges_with_analysis()`:

1. **Single pass:** Stream edges through `scan_edges_with_analysis()` to simultaneously count degrees, classify core nodes, and collect core-induced edges. Returns `(in_deg, out_deg, is_core, directed_core_edges)`.
2. **Prism detection:** Build CSR for the core-induced subgraph, run $O(N)$ 2-hop Causal Prism detection and generation classification.
3. **Spectral measurement:** Run random walkers on the core CSR for spectral dimension; build directed CSR for causal flux.

4.3 DefectResult Struct

Field	Description
<code>vac_head, vac_data</code>	Vacuum Hasse CSR (row pointers + column indices).
<code>def_head, def_data</code>	Defect CSR after threat contraction + Prism pole-completion edges.
<code>vacuum_core</code>	Core node indices (top 10% by degree) in the vacuum graph.
<code>defect_core</code>	Core node indices surviving threat contraction.
<code>gen1_nodes</code>	Generation 1 Prism node indices — most abundant signature (electron-like).
<code>gen2_nodes</code>	Generation 2 Prism node indices — second most abundant (muon-like).
<code>gen3_nodes</code>	Generation 3 Prism node indices — third most abundant (tau-like).
<code>anti1_nodes</code>	Anti-Generation 1 Prism node indices — CPT conjugate (positron-like).
<code>sterile_nodes</code>	Sterile prism node indices — prisms with $\Phi = 0$ (fully phase-cancelled, dark matter; Phase-Coherence Theorem).
<code>merge_into</code>	Merge map: <code>merge_into[i] = canonical node for node i</code> . For threat-absorbed nodes, points to the absorbing Prism pole.

Table 4.1: DefectResult fields.

4.4 TopologySummary Struct

Field	Description
total_nodes	Total causet cardinality N .
total_prisms	Number of committed Causal Prisms detected.
max_intermediates	Largest belly size N among all prisms.
count_gen1..3	Number of prisms in each generation.
count_antigen1	Number of antimatter prisms (CPT conjugate of Gen 1).
count_sterile	Number of sterile prisms ($\Phi = 0$, fully phase-cancelled).
avg_mass_gen1..3	Average topological mass $\langle N \rangle$ per generation.
avg_mass_sterile	Average topological mass for sterile prisms.
visible_mass_total	$\sum \Phi(P) $ — total visible (EM) mass across all prisms.
dark_mass_total	$\sum (N - \Phi(P))$ — total dark mass across all prisms.
grav_mass_total	$\sum N$ — total gravitational mass (identity: = vis + dark).
omega_ratio	$\Omega_{\text{linear}} = \sum (N - \Phi) / \sum \Phi $ — linear mass ratio (graph-theoretic identity).
phase_sq_total	$\sum \Phi(P) ^2$ — total EM self-energy (numerator of $\mathcal{Q}_{\text{topo}}$).
mass_sq_total	$\sum N^2$ — total gravitational self-energy (denominator of $\mathcal{Q}_{\text{topo}}$).
alpha_em	$\alpha = \mathcal{Q}_{\text{topo}} / (8\pi)$ — emergent fine structure constant, where $\mathcal{Q}_{\text{topo}} = \sum \Phi ^2 / \sum N^2$.
omega_energy	$\Omega_{\text{energy}} = 1 / \mathcal{Q}_{\text{topo}} - 1$ — self-energy dark matter ratio. Satisfies the exact identity $\alpha(1 + \Omega_{\text{energy}}) = 1/(8\pi)$.
phase_pos_count	Count of intermediate nodes with $\varphi = +1$ (source-like, out-degree > in-degree).
phase_zero_count	Count of intermediate nodes with $\varphi = 0$ (balanced in/out degree).
phase_neg_count	Count of intermediate nodes with $\varphi = -1$ (sink-like, in-degree > out-degree).
prisms_gen1..3	Number of <i>prisms</i> (not nodes) per generation.

Table 4.2: TopologySummary fields. The Phase-Coherence Mass Decomposition ($M_{\text{vis}} = |\Phi|$, $M_{\text{dark}} = N - |\Phi|$) replaces the former Conjecture C6 threshold ($N > 5$) with a parameter-free identity. The self-energy ratio Ω_{energy} and the fine structure constant α are algebraically locked by $\alpha(1 + \Omega) = 1/(8\pi)$. The phase census fields support the occupancy model analysis (see §10.4).

4.5 Signature Encoding (Legacy Diagnostic)

Note: The primary generation classification now uses the Vol I phase function ($g(P)$, $\Phi(P)$) described above. The quartile signature encoding below is retained for diagnostic logging only.

Each Causal Prism signature is computed from the bulk momentum of ALL components (origin, destination, and intermediates):

1. Collect bulk momentum $[M_{\text{origin}}, M_{\text{destination}}, M_{w_1}, M_{w_2}, \dots]$ for all nodes in the Prism.
2. Sort the list and extract quartile values: $[q_0, q_1, q_2, q_3]$ (0th, 25th, 50th, 75th percentiles).
3. Normalise each quartile to $[-10, +10]$ via $q_i \times 10 / \max(|q|)$.

4. Offset by +16 to make non-negative (range $[6, 26]$, fits in 5 bits).
5. Pack four 5-bit components into a single u32:

$$\text{sig} = c_0 \mid (c_1 \ll 5) \mid (c_2 \ll 10) \mid (c_3 \ll 15)$$

The anti-signature (CPT conjugate) is computed by reversing and negating: $\bar{c}_i = -c_{3-i} + 16$.

Module Reference: **spectral** — Phase 3

Spectral Dimension Measurement

- *Random walk return probability via undirected and directed walkers.*
- *Two tiers: eigendecomposition (exact, $N \leq 3k$ by default; adjustable via `--eigen-cutoff`) and Monte Carlo (above the cutoff).*
- *Per-generation and causal flux measurements.*

5.1 Functions

5.1.1 `spectral_dimension(steps, p_vals) → Vec<f64>`

$d_S(t) = -2 d(\ln P)/d(\ln t)$ via centred finite differences. Forward difference at $t = 1$, backward at $t = t_{\max}$, centred elsewhere.

5.1.2 `run_walkers(adj_head, adj_data, starts, steps, seed) → Vec<f64>`

Run W independent lazy-walk walkers in parallel. At each step: 50% stay, 50% move to a uniformly random neighbour. Returns $P(t)$ = fraction of walkers that returned to their starting node at each step. Parallel via Rayon.

Strict Finitism: all internal accumulation uses u64 integer arithmetic. Each walker contributes exactly 0 or 1 (Kronecker delta) per measurement step. The reduction sums u64 counts across all walkers. The single f64 division count / W occurs only at the final return, preserving exact integer semantics throughout the Monte Carlo core.

5.1.3 `run_transmission_walkers(...) → (Vec<f64>, Vec<f64>)`

Directed walkers (mandatory move along causal arrow). Returns two vectors: attraction flux (Gen1 → AntiGen1) and repulsion flux (Gen1 → Gen1). Walkers that reach a dead end (edge of causal set) escape and stop. Same u64 integer accumulation as `run_walkers`.

5.1.4 `compute_eigen(n, rows, cols, steps, core) → SpectralResult`

Exact $P(t) = N^{-1} \sum_k \lambda_k^t$ from eigendecomposition of the symmetric normalised adjacency $S = D^{-1/2} A D^{-1/2}$. Complexity: $O(N^3)$. Used only for small graphs ($N \leq 3k$). Local $P(t)$ uses eigenvector-weighted sums over core indices.

5.1.5 `compute_monte_carlo(n, rows, cols, steps, core, n_w, rng) → SpectralResult`

Monte Carlo $P(t)$ from random walkers. Builds CSR adjacency internally, then launches lazy random walkers. Complexity: $O(W \cdot t)$, independent of N .

5.1.6 `compute_monte_carlo_csr(n, head, data, steps, core, n_w, rng) → SpectralResult`

Zero-copy variant — avoids rebuilding adjacency when CSR is already available from Phase 2. This is the hot path for large N in-memory runs.

5.1.7 `build_adj_list(n, rows, cols) → (Vec<u32>, Vec<u32>)`

Build sorted undirected adjacency list in CSR format from edge lists.

5.1.8 make_symmetric(n, head, data) → (Vec<u32>, Vec<u32>)

Convert a directed CSR (forward-only Hasse DAG) into an undirected adjacency list. Used to symmetrize the vacuum CSR before running spectral walkers, since the Hasse diagram stores only forward (past→future) edges but random walkers need to step in both directions.

5.2 SpectralResult Struct

Field	Description
p_global	$P(t)$ from uniformly random starts. Probes global geometry.
ds_global	$d_S(t)$ of the bulk (derived from p_global).
p_local	$P(t)$ from walkers starting on core nodes.
ds_local	$d_S(t)$ from core walkers. Sensitive to topological defects.
p_gen1, ds_gen1	$P(t)$ and d_S for Generation 1 Causal Prism nodes (electron-like).
p_gen2, ds_gen2	$P(t)$ and d_S for Generation 2 Prism nodes (muon-like).
p_gen3, ds_gen3	$P(t)$ and d_S for Generation 3 Prism nodes (tau-like).
p_anti1, ds_anti1	$P(t)$ and d_S for Anti-Generation 1 (positron-like, CPT conjugate).
flux_attraction	Directed transmission: Gen1 → AntiGen1 (opposite charge).
flux_repulsion	Directed transmission: Gen1 → Gen1 (same charge).
flux_attr_norm	Normalized attraction flux: $\text{flux_attraction} / \text{targets} $. Isolates intrinsic per-node coupling strength (per-charge definition, Vol II).
flux_repu_norm	Normalized repulsion flux: $\text{flux_repulsion} / \text{targets} $.
p_sterile, ds_sterile	$P(t)$ and d_S for walkers on sterile prism nodes ($\Phi = 0$). Phase-Coherence Theorem: gravitationally active, electromagnetically silent (dark matter).
ds*_std	Population standard deviation of each $d_S(t)$ field across M realisations. Empty when $M = 1$. Provides error bars for publication figures.
flux*_std	Population standard deviation of each flux field across M realisations.

Table 5.1: SpectralResult fields (28 total).

Module Reference: `anim_export` — Topology Slice Export

Binary Export for External Renderers

- *Pre-computed binary topology data for keyframe playback (e.g. CausalAnim).*
- *Invoked via `--export-slice <PATH>`: runs Phases 1–2 only, no spectral walkers.*
- *Output format: bincode-serialized `TopologySlice` containing coordinates, directed Hasse edges, and Causal Prisms.*

6.1 Structs

6.1.1 TopologySlice

The top-level serializable structure written to the `.slice` file:

Field	Description
<code>n_total</code>	Number of spacetime events.
<code>coordinates</code>	<code>Vec<f64; 4></code> — 4D coordinates (time-sorted) for all events.
<code>hasse_edges</code>	<code>Vec<(u32, u32)></code> — directed Hasse links (past→future), filtered by time ordering $t_u < t_v$.
<code>prisms</code>	<code>Vec<PrismDef></code> — all committed Causal Prisms from Phase 2.

6.1.2 PrismDef

Lightweight u32-indexed Causal Prism definition (converted from internal `CausalPrism` with usize indices):

Field	Description
<code>origin</code>	Past pole index (u32).
<code>destination</code>	Future pole index (u32).
<code>intermediates</code>	<code>Vec<u32></code> — belly node indices.

6.2 Function

6.2.1 `write_slice(path, slice) → Result<(), String>`

Serializes a `TopologySlice` to bincode format and writes the bytes to the given file path. At $N = 50,000$ the output file is typically ~ 5 MB; at $N = 5,000$ it is ~ 800 KB.

6.3 Usage

The `--export-slice` flag triggers an early-exit branch in `main.rs`:

1. Run Phase 1: `sprinkle + build_hasse_sparse` or `build_hasse_direct`.
2. Run Phase 2: `apply_defect` → `DefectResult + Vec<CausalPrism>`.
3. Extract directed Hasse edges from the vacuum CSR (filter by $t_u < t_v$).

4. Convert `CausalPrism` $\rightarrow$ `PrismDef` (`usize` $\rightarrow$ `u32`).
5. Serialize `TopologySlice` via `write_slice` and exit.

Phases 3 and 4 (spectral walkers, ensemble averaging, CSV output) are skipped entirely, making the export fast even for large N .

Module Reference: **output** — Phase 4

CSV Serialisation

- *Ensemble-averaged observables written to CSV (results.csv).*
- *32-column format: return probabilities, spectral dimensions, raw and normalized causal flux, sterile prism observables ($\Phi = 0$), mass spectrum, and ensemble standard deviations (error bars for $M > 1$ runs).*
- *Topology summary (topology_summary.csv): prism census, generation abundances, Phase-Coherence Mass Decomposition ($\Omega_d/\Omega_v, \alpha$).*
- *Mass spectrum histogram (mass_spectrum.csv): belly size N vs frequency.*

7.1 Functions

7.1.1 write_csv(path, steps, vacuum, defect, metadata)

Serialise vacuum and defect spectral results to a CSV file. Each row corresponds to one diffusion step t . The vacuum result provides control measurements (unmodified Hasse graph); the defect result provides measurements after Kuratowski contraction (Causal Prism + K_5 threat absorption).

7.1.2 export_topology_summary(path, topo, metadata)

Export a single-record key-value CSV containing the structural fingerprint of the generated universe: prism counts, generation abundances, average topological masses, and the Phase-Coherence Mass Decomposition fields (visible_mass_total, dark_mass_total, grav_mass_total, omega_ratio, phase_sq_total, mass_sq_total, q_topo, alpha_em, omega_energy).

7.1.3 export_mass_spectrum(path, histogram, metadata)

Export the prism mass spectrum histogram as a two-column CSV (intermediates_N, frequency). Each row maps a belly size N to the number of committed prisms with that exact belly size.

7.2 CSV Column Reference

Column	Physics	Units
step	Diffusion time t	integer steps
P_vac	Return probability on vacuum graph	probability
dS_vac	Spectral dimension of vacuum	dimensionless
P_def	Return probability on defect graph	probability
dS_def	Spectral dimension with matter	dimensionless
P_Gen1	Return probability for Generation 1	probability
dS_Gen1	Spectral dimension, Generation 1	dimensionless
P_Gen2	Return probability for Generation 2	probability
dS_Gen2	Spectral dimension, Generation 2	dimensionless
P_Gen3	Return probability for Generation 3	probability
dS_Gen3	Spectral dimension, Generation 3	dimensionless
P_Anti1	Return probability for antimatter	probability
dS_Anti1	Spectral dimension of antimatter	dimensionless
Flux_Attr	Causal flux: Gen1→AntiGen1 (attraction)	probability
Flux_Repu	Causal flux: Gen1→Gen1 (repulsion)	probability
Flux_Attr_Norm	Normalized flux: Attr / targets (per-charge def.)	per-node prob
Flux_Repu_Norm	Normalized flux: Repu / targets (per-charge def.)	per-node prob
P_Sterile	Return prob. for sterile prisms ($\Phi = 0$)	probability
dS_Sterile	Spectral dimension of sterile prisms (Phase-Coherence Thm)	dimensionless
Mass_Gen1..3	Avg topological mass per generation	integer (avg)
Mass_Anti1	Avg topological mass for antimatter	integer (avg)
dS_vac_std	Std dev of d_S vacuum across realisations	dimensionless
dS_def_std	Std dev of d_S defect across realisations	dimensionless
dS_Gen1..3_std	Std dev of d_S per generation	dimensionless
dS_Anti1_std	Std dev of d_S antimatter	dimensionless
dS_Sterile_std	Std dev of d_S sterile	dimensionless
Flux_Attr_std	Std dev of flux attraction	probability
Flux_Repu_std	Std dev of flux repulsion	probability

Table 7.1: CSV output columns (32 total).

Module Reference: **memory** — RAM-Aware Mode Selection

Chapter Overview

- Automatic detection of available system RAM.
- Memory estimation heuristics for each simulation tier.
- Interactive mode selection with config file override.

8.1 Functions

8.1.1 `load_config(dir) → Config`

Load configuration from a TOML file, or return defaults. Searches for `causal_set.toml` in the given directory first, then falls back to the current working directory.

8.1.2 `ensure_config(dir)`

Write the default config file if it doesn't already exist.

8.1.3 `recommend_mode(n, cfg) → (ExecMode, u64, u64)`

Recommend an execution mode based on config + available RAM vs estimated usage. The estimate accounts for concurrent realisations (default: 2 for $N > 10\text{M}$, 4 otherwise; overridable via `--threads`).

8.1.4 `prompt_mode(recommended, available, estimated, cfg) → ExecMode`

Prompt the user to confirm or override the recommended execution mode.

8.1.5 `estimate_memory_bytes(n) → u64`

Heuristic peak memory estimate for in-memory mode:

- $N \leq 3,000$: $\sim N^2 \times 24$ bytes (dense eigendecomp matrices).
- $3,000 < N \leq 50,000$: $\sim N \times 2,000$ bytes (sparse structures).
- $N > 50,000$: $\sim N \times 500$ bytes (OCI + CSR + defect).

8.1.6 `estimate_streaming_memory_bytes(n) → u64`

Peak memory estimate for streaming mode: $13N + 100\text{ MB}$. Components: $8N$ (degree arrays) $+ 50 \cdot (N/10)$ (core CSR) $+ 100\text{ MB}$ (buffer). At $N = 100\text{M}$: $\sim 1.3\text{ GB}$ per realisation.

8.1.7 `max_concurrent_runs(n) → usize`

Default concurrency heuristic: 2 for $N > 10\text{M}$, 4 otherwise. Used as the fallback when `--threads` is not provided.

8.1.8 `available_ram_bytes() → u64`

Detect available system RAM in bytes via the `sysinfo` crate.

8.2 Types

Definition 1: ExecMode

An enum: InMemory (keep all data in RAM) or Streaming (single-pass edge analysis via `scan_edges_with_analysis()`, ~ 1.3 GB per realisation at $N = 100M$).

Definition 2: Config

User-configurable settings: `max_ram_bytes` (hard ceiling, `0 = auto`), `safety_fraction` (0.0–1.0), `output_dir` (default output directory).

Usage

9.1 Options

Argument	Description	Default
<code>N</code>	Number of spacetime events	5000
<code>M</code>	Ensemble realisations	10
<code>output_dir</code>	Directory for results	current dir
<code>--stream</code>	Force streaming mode	—
<code>--inmemory</code>	Force in-memory mode	—
<code>--seed <i>u64</i></code>	Base seed (each realisation uses seed+i)	system clock
<code>--threads <i>usize</i></code>	Max parallel realisations	auto (2–8)
<code>--epsilon <i>f64</i></code>	Topological error tolerance ε	0.01
<code>--tmax <i>usize</i></code>	Maximum diffusion time	15
<code>--eigen-cutoff <i>N</i></code>	Eigendecomp threshold	3000
<code>--target-error <i>f64</i></code>	Relative SE threshold for adaptive convergence	0.05
<code>--max-ensemble <i>usize</i></code>	Maximum realisations (hard cap)	22
<code>--min-ensemble <i>usize</i></code>	Minimum realisations before convergence	3
<code>--batch-size <i>usize</i></code>	Realisations per parallel batch	auto
<code>--resume</code>	Resume from checkpoint	—
<code>--export-slice <i>PATH</i></code>	Export topology slice (single realization, no ensemble)	—
<code>--help</code>	Show help message	—

9.2 Configuration

Place a `causal_set.toml` in the working directory (auto-generated on first run if absent):

```
# Hard limit on RAM (GB). 0 = auto-detect from system.
max_ram_gb = 0

# Fraction of available RAM considered safe (when auto-detecting).
safety_fraction = 0.70

# Default output directory for results and streaming edge files.
output_dir = "."
```

9.3 Examples

```
# Standard in-memory run
cargo run --release -- 50000 10

# Streaming run (large N, single-pass O(N))
cargo run --release -- 1000000 5 --stream

# Reproducible run with explicit seed
cargo run --release -- 50000 10 --inmemory --seed 12345

# 16 parallel realisations on a high-RAM workstation
```

```
cargo run --release -- 10000000 20 --inmemory --threads 16

# Force eigendecomp up to N=5000 for exact comparison
cargo run --release -- 5000 10 --inmemory --eigen-cutoff 5000

# Custom walker precision (fewer walkers for exploratory runs)
cargo run --release -- 50000 10 --inmemory --epsilon 0.05 --tmax 10

# Custom output directory
cargo run --release -- 50000 10 /path/to/output

# Export topology slice for external renderers (Phases 1-2 only)
cargo run --release -- 50000 --inmemory --seed 42 \
  --export-slice /tmp/universe.slice
```

Expected Results

10.1 Spectral Dimension

Production ensemble: $N = 10^7$, $M = 20$ realisations, seed 42, runtime ~ 4.5 hours on a ThinkPad T480 (Intel i5-8250U, 16 GB RAM).

- **Vacuum (UV, $\sigma=1$):** $d_S = 1.949 \pm 0.003$ (UV dimensional reduction $d_S \rightarrow 2$).
- **Vacuum (IR, $\sigma=4$):** $d_S = 4.956 \pm 0.001$ (4D Minkowski continuum emerges).
- **Defect (core, peak):** $d_S \approx 9.06$ (topological retention — the Prism bipartite structure traps probability flow and elevates the effective dimension).

Key Result

Mass is not a continuous parameter assigned to a particle, but a quantized topological property: the integer count N of intermediate nodes in the Causal Prism. This topological inertia stabilises the spectral dimension at a finite value, while the vacuum diverges.

10.2 Generation Classification

Causal Prisms are classified by the Vol I phase function $\varphi(w) = \text{sign}(\text{bulk_momentum}[w])$:

Generation	Physics Analogue	Classification	Mass ($N=10^7$, $M=20$)
Gen1	Electron-like	$g=1 \wedge \Phi \geq 0$	4.555 (lightest)
Gen2	Muon-like	$g=2$	6.531 (medium)
Gen3	Tau-like	$g=3$	7.732 (heaviest)
AntiGen1	Positron-like	$g=1 \wedge \Phi < 0$	4.45 (CPT, $\Delta m/m = 2.3\%$)

At $N = 10^7$ ($M=20$): Total prisms: 7,723,943. $g=1$: 63,278 (2.1%), $g=2$: 2,552,871 (84.9%), $g=3$: 391,039 (13.0%). Sterile ($\Phi=0$): 400,986. Mass ratios 1 : 1.43 : 1.70 are N -independent topological invariants.

10.3 Coupling Constants

Observable	$N=10^7$	$N \rightarrow \infty$	Note
$\mathcal{Q}_{\text{topo}}$	0.191	0.152 ± 0.001 ($R^2=0.9996$)	Topological charge ratio
α^{-1}	131.8	165.1 ± 1.0	Bare coupling at Planck scale
Ω_{energy}	4.24	5.57 (Planck 2018: 5.36)	Self-energy dark matter ratio
$\alpha(1 + \Omega)$	$1/(8\pi)$	$1/(8\pi)$	Exact at every N

10.4 Occupancy Model (Analytical Mass Hierarchy)

The mass hierarchy $m_1 < m_2 < m_3$ is explained analytically as a **coupon-collector selection effect** on the belly size distribution $f(N)$. Each intermediate node w carries a causal phase $\varphi(w) = \text{sign}(\text{out_degree} - \text{in_degree}) \in \{-1, 0, +1\}$. The generation $g(P)$ counts how many distinct phase signs appear among the N intermediates of prism P .

Since $P(g=1 | N) \sim p_{\max}^N$ decays exponentially, Gen 1 prisms are biased toward small bellies (low mass), while Gen 3 prisms require all three signs (large bellies, high mass). The expected mass per generation is

$$E[N | g=k] = \frac{\sum_N N \cdot P(g=k | N) \cdot f(N)}{\sum_N P(g=k | N) \cdot f(N)},$$

where the generation probabilities follow from inclusion-exclusion:

$$\begin{aligned} P(g=1 | N) &= p_+^N + p_0^N + p_-^N, \\ P(g=3 | N) &= 1 - (1-p_+)^N - (1-p_0)^N - (1-p_-)^N + p_+^N + p_0^N + p_-^N. \end{aligned}$$

The simulation reports the **intermediate phase census**: $(p_+, p_0, p_-) = (0.318, 0.018, 0.664)$. Plugging these measured fractions into the occupancy formula reproduces the mass ratios with **zero free parameters**:

Ratio	Predicted	Observed	Error
m_2/m_1	1.409	1.434	1.8%
m_3/m_1	1.674	1.698	1.4%

Table 10.1: Occupancy model predictions versus simulation. Both inputs—the belly distribution $f(N)$ and the phase fractions—are determined entirely by the Poisson sprinkling geometry. Analysis script: `data/scripts/occupancy_model.py`.

10.5 Causal Flux

Directed walker transmission probabilities measure electromagnetic-like interactions:

- **Attraction:** Gen1 $\rightarrow$ AntiGen1 (opposite charge, positive flux).
- **Repulsion:** Gen1 $\rightarrow$ Gen1 (same charge, lower transmission).

Dependencies

Crate	Purpose
<code>nalgebra</code>	Dense eigendecomposition (Tier 1, $N \leq \text{cutoff}$; default 3k)
<code>sprs</code>	Sparse matrix algebra for A^2 Hasse reduction
<code>rand</code>	Deterministic PRNG (seeded <code>StdRng</code>)
<code>rayon</code>	Data-parallel iteration across walkers and nodes
<code>sysinfo</code>	System RAM detection for automatic mode selection
<code>smallvec</code>	Stack-allocated small vectors for inner-loop optimisation
<code>serde + bincode</code>	Serialization for checkpoint files and <code>.slice</code> topology export

Table 11.1: Rust crate dependencies.

See Also

- *Modulo Synthesis Vol I: The Geometry of Time* — Axiom 1, causal sets, spectral dimension flow, the Geometry of Logic ($O(N)$ Locality Theorem), Law of Large Numbers.
- *Modulo Synthesis Vol II: The Geometry of Matter* — Axiom 2, Kuratowski's Theorem, generation theorem, colour confinement, gauge forces, the Three Geometries.
- *Emma's Thought Experiments* — Pedagogic companion with intuitive explanations, dimensional adjustment, and locality.

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*“The universe is strictly computable, finite, and structurally homeostatic.
The rest is just counting.”*